

Skills Summary

End Behavior and Multiplicity

Use this reference while completing your worksheet or when studying.

Skill 1 — Zeros and what their multiplicity does

Rule. A factor $(x - r)^m$ gives a zero at $x = r$ of multiplicity m .

- m odd: the curve *crosses* the axis at r (the product changes sign)
- m even: the curve *touches* and turns back (no sign change)

The multiplicities add up to the degree.

Example: List the zeros of $p(x) = (x - 4)(x + 1)^2$.

Step 1. Each factor contributes one zero, and its exponent is what says whether the curve crosses there — so read the exponents, not just the roots.

Step 2. $x - 4 = 0$ gives a crossing at 4; $(x + 1)^2 = 0$ gives a touch at -1 .

$$\Rightarrow x = 4 \text{ and } x = -1$$

Try it: List the zeros of $p(x) = (x + 5)(x - 2)^2$, and say which is a touch.

check: $x = -5$ crossing, $x = 2$ touch

Skill 2 — End behavior from the leading term

Rule. Only the degree and the sign of the leading coefficient matter.

- degree *even*: both ends go the same way — up if the leading coefficient is positive, down if negative
- degree *odd*: the ends go opposite ways — up on the right if positive, down on the right if negative

Example: Find $\lim_{x \rightarrow \infty} (-4x^3 + x)$.

Step 1. Every other term is out-grown by the leading one, so discard them and read off the sign of $-4x^3$.

Step 2. For large positive x , x^3 is large and positive, and multiplying by -4 makes it large and negative.

$$\Rightarrow -\infty$$

Try it: Find $\lim_{x \rightarrow -\infty} (2x^6 - 9x)$.

check: ∞

(!) Watch out: the constant term never affects end behavior, and neither does any middle term. It shifts the curve but cannot change where the two ends go.

Skill 3 — Assembling the sketch

Four steps. Plot the zeros and mark each as a crossing or a touch; find the y -intercept by evaluating at 0; draw the two ends from the degree and the leading sign; join everything with one smooth curve that changes sign only where it should.

Example: Find the y -intercept of $p(x) = (x - 3)(x + 1)^2$, then describe the sketch.

Step 1. The intercept is the one point not given by the factors, so evaluate at 0 before drawing anything.

Step 2. $p(0) = (0 - 3)(0 + 1)^2 = -3$; the curve crosses at 3, touches at -1 , and being degree 3 with a positive lead it falls left and rises right.

$$\Rightarrow p(0) = -3$$

Try it: Find the y -intercept of $p(x) = (x - 2)(x + 4)^2$.

check: $p(0) = -8$